

□ Recalls, Preliminaries

1. Restriction thm.

For a Poisson process Π with state space S and mean measure μ ,

for a subset $S_1 \subset S$, Let $\Pi_1 := \Pi \cap S_1$,

Then Π_1 can be regarded as

① Poisson process on S with mean measure $\mu^*(A) := \mu(A \cap S_1)$

② " " " S_1 " " " $\mu|_{S_1}$

2. Mapping thm.

For a Poisson process Π with σ -finite mean measure μ on S

and $f: S \rightarrow T$: measurable fn s.t induced measure $\mu^*(B) = \mu(f^{-1}(B))$ $B \subset T$

has no atom, then $f(\Pi)$ is a Poisson process on T with mean measure μ^*

3. Characteristic functional

For a random subset Π of $\mathbb{R}$, let $\Sigma_f := \sum_{x \in \Pi} f(x)$

If the random variable Σ_f satisfies

$$E[e^{-\Sigma_f}] = \exp \left\{ - \int_{-\infty}^{\infty} (1 - e^{-f(x)}) \lambda dx \right\}$$

for all positive step function f , then Π is the Poisson process with rate λ .

① Properties of the Poisson process on $\mathbb{R}$

In this Chapter. let

π : Poisson process on $\mathbb{R}$ with constant rate λ .

$$\text{i.e. } \forall A \subset \mathbb{R}, \quad P(N(A) = n) = \pi_n(\lambda|A|) = \frac{(\lambda|A|)^n e^{-\lambda|A|}}{n!}$$

Then we have that

$$\textcircled{1} \text{ For any bounded set } A \subset \mathbb{R}, \quad P(N(A) < \infty) = 1$$

$$\textcircled{2} P(N(0, \infty) = \infty) = P(N(-\infty, 0) = \infty) = 1.$$

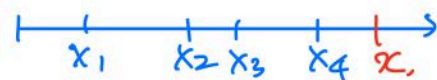
$$\textcircled{3} P(0 \in \pi) = P(N(\{0\}) \geq 1) = 0$$

$\Rightarrow$ The points of π are not bounded, and don't have the accumulation point.

Hence, $\pi \cap (0, \infty)$ can be written as $\{x_1, x_2, \dots\}$ with $0 < x_1 < x_2 < \dots$

Also $\pi \cap (-\infty, 0)$ " " " " $\{\dots, x_{-2}, x_{-1}\}$ with $\dots < x_{-2} < x_{-1} < 0$

Remark X_n is a random variable.



$$p.f.) \{ \omega \in \Omega \mid X_n(\omega) \leq x \} = \{ \omega \in \Omega \mid N(0, x] \geq n \} \in \mathcal{F} \quad x_n \leq x \Leftrightarrow N(0, x] \geq n$$

Hence π can be regarded as the sequence of the random variable.

$\Rightarrow$ Unique property for $S = \mathbb{R}$ case.

By Restriction thm, $\pi \cap (0, \infty)$ and $\pi \cap (-\infty, 0)$ are the Poisson process.

By Mapping thm, the joint distribution of $\pi \cap (0, \infty)$ and $\pi \cap (-\infty, 0)$ are same.

$\Rightarrow$ We will consider the $\pi \cap (0, \infty) = (x_1, x_2, \dots)$ only.

The following theorem describes the structure of $(x_1, x_2, x_3, \dots)$

Thm (Interval thm)

The r.v. $Y_1 = X_1$, $Y_n = X_n - X_{n-1}$ ($n \geq 2$) are independent,

each follows the exponential distribution $\text{Exp}(\lambda)$

$$\text{i.e. } f_{Y_n}(y) = \lambda e^{-\lambda y} \quad (y > 0)$$

pf) Since the independence between the random set and random variable

is not well-defined in books, I will not use that term.

The main idea is using the version of the strong markov property.

→ If the origin moved to the random point X_1 ,

the points of Π to the right of X_1 form a new Poisson process

that is independent of X_1 . (Scheme)

Note: Mapping that can't be applied since $x \mapsto x - X_1$

is not deterministic. → use characteristic functional.

$$\text{Let } \Pi^1 := \{X_2 - X_1, X_3 - X_1, \dots\}$$

$$\Sigma_f^1 := \sum_{n \geq 2} f(X_n - X_1), \quad \Sigma_f := \sum_{n=1} f(X_n)$$

for $\begin{cases} f: \text{positive left-continuous} \\ \text{step function} \\ \text{with } f(x) = 0 \quad \forall x \leq 0 \\ \text{and } f \text{ vanishes outside} \\ \text{the bdd set.} \end{cases}$

Claim Σ_f^1 has the same distribution with Σ_f , Σ_f^1 is independent with X_1

pf) let $\varphi_k :=$ least integer multiple of X_1 greater than $x' \in \text{r.v.}$

$$\text{let } \Sigma_f^{1,k} := \sum_{n \geq 2} f(X_n - \varphi_k)$$

$$\Rightarrow P(\sum_f^{l,k} \leq z, X_1 \leq x) = \sum_{l \geq 1} P(\sum_f^{l,k} \leq z, X_1 \leq x, \xi_k = l \cdot 2^{-k}) \quad (\text{law of total prob.})$$

and by fixed l , $\Pi \cap (l2^{-k}, \infty)$ form a Poisson process (Restriction)

$\Rightarrow$ with mapping $x \mapsto x - l2^{-k}$, $\{X_2 - l2^{-k}, X_3 - l2^{-k}, \dots\} \cap (0, \infty)$: Poisson process

$$\Rightarrow \text{so with } \xi_k = l2^{-k} \text{ given, } \sum_f^{l,k} = \sum f(X_n - l2^{-k})$$

has the same distribution with $\sum_f$ don't affect

and independent with X_1 ($\because X_1 < l2^{-k} \rightarrow f(X_1 - l2^{-k}) = 0$ the $\sum_f^{l,k}$)

$$\therefore P(\sum_f^{l,k} \leq z, X_1 \leq x, \xi_k = l2^{-k}) = P(\sum_f \leq z) P(X_1 \leq x, \xi_k = l2^{-k})$$

$$\Rightarrow P(\sum_f^{l,k} \leq z, X_1 \leq x) = P(\sum_f \leq z) P(X_1 \leq x)$$

$$= P(\sum_f \leq z) P(N(0, x] \geq 1)$$

$$= P(\sum_f \leq z) (1 - e^{-\lambda x})$$

and since f is left continuous, and $\sum_f^{l,k}, \sum_f$ are both finite sum

with probability 1, and $\xi_k \xrightarrow{a.s.} X_1$, $\sum_f^{l,k} \xrightarrow{a.s.} \sum_f$

$$\Rightarrow \text{letting } k \rightarrow \infty \text{ gives } P(\sum_f \leq z, X_1 \leq x) = P(\sum_f \leq z) (1 - e^{-\lambda x})$$

This gives the independence between $\sum_f$ and X_1 , and $F_{X_1}(x) = 1 - e^{-\lambda x}$

letting f :

$$\{ \sum_f^1 = 1 \} = \{ X_2 - X_1 \leq x_2 \}$$

$\Rightarrow (X_2 - X_1)$ and X_1 are independent.

Use induction to deduce the independency between

$$\{X_1, X_2 - X_1, \dots, X_n - X_{n-1}\} \text{ and } \sum_f^n := \sum_{m=n+1}^{\infty} f(X_m - X_n)$$

$\Rightarrow$ Indep. between X_1 & $X_{n+1} - X_n, X_2 - X_1$ & $X_{n+1} - X_n, \dots, X_n - X_{n-1}$ & $X_{n+1} - X_n$

Then finally,

$$\begin{aligned} & P(X_{n+1} - X_n \leq y_{n+1}) \\ &= \int_0^\infty \int_0^\infty \cdots \int_0^\infty \underbrace{P(X_{n+1} - X_n \leq y_{n+1} \mid Y_1 = y_1, Y_2 = y_2, \dots, Y_n = y_n)}_{= P(N(y_1 + \dots + y_n, y_1 + \dots + y_{n+1}) \geq 1)} f_{Y_1}(y_1) f_{Y_2}(y_2) \cdots f_{Y_n}(y_n) dy_1 \cdots dy_n \\ &= (1 - e^{-\lambda y_{n+1}}) : \text{ftr of } y_{n+1} \text{ only.} \\ &= (1 - e^{-\lambda y_{n+1}}) \end{aligned}$$

2 Law of Large Numbers.

By Interval thm, $X_n = Y_1 + Y_2 + \dots + Y_n$ where $Y_i \stackrel{i.i.d}{\sim} \text{Exp}(\lambda)$

$$\Rightarrow E[Y_i] = \frac{1}{\lambda}, \text{Var}(Y_i) = \frac{1}{\lambda^2}, \Rightarrow E[X_n] = \frac{n}{\lambda}, \text{Var}(X_n) = \frac{n}{\lambda^2}$$

By the Strong law of Large number, by regarding $\frac{X_n}{n}$ as the sample mean calculated from the data $Y_1, Y_2, \dots, Y_n$, we have

$$P\left(\lim_{n \rightarrow \infty} \frac{X_n}{n} = \frac{1}{\lambda}\right) = 1$$

$$\text{Since } N(0, t] \rightarrow \infty \text{ as } t \rightarrow \infty, \quad P\left(\lim_{t \rightarrow \infty} \frac{X_{(0, t]}}{t} = \frac{1}{\lambda}\right) = 1.$$

Since $X_{N(0, t]} < t < X_{N(0, t]+1}$ holds for every t , by sandwich

$$\text{we have that } P\left(\lim_{t \rightarrow \infty} \frac{t}{N(0, t]} = \frac{1}{\lambda}\right) = 1$$

$$\Rightarrow P\left(\lim_{t \rightarrow \infty} \frac{N(0, t]}{t} = \lambda\right) = 1.$$

We will suggest the direct proof that does not use the interval thm.

pf) let $Z_m = N(0, m) \Rightarrow Z_m \sim P(\lambda m) \Rightarrow E[Z_m] = \lambda m, \text{Var}(Z_m) = \lambda m$

Thm (Chebyshev's inequality)

for r.v. X with $E[X] = \mu, \text{Var}(X) = \sigma^2, P(|X - \mu| \geq k) \leq \frac{\sigma^2}{k^2}$

By Chebyshev, $P(|\frac{Z_m}{m} - \lambda| \geq \varepsilon) \leq \frac{\lambda}{\varepsilon^2 m}$

take $m = k^2$, then $P(|\frac{Z_{k^2}}{k^2} - \lambda| \geq \varepsilon) = \frac{\lambda}{\varepsilon^2} \cdot \frac{1}{k^2}$

Thm (Borel-Cantelli)

Let $A_1, A_2, \dots$ ^{msble} events, $\limsup A_n := \bigcap_{n=1}^{\infty} \bigcup_{m=n}^{\infty} A_m$: " A_n infinitely often occur"

If $\sum P(A_n) < \infty$, then $P(\limsup A_n) = 0$

let the Event $A_k := \{|\frac{Z_{k^2}}{k^2} - \lambda| \geq \varepsilon\} \Rightarrow \sum P(A_k) < \infty$

$\Rightarrow$ By Borel-Cantelli, $P(\limsup A_k) = 0$

$\Rightarrow \{A_k \text{ occurs infinitely often}\}$ has probability 0

$\Rightarrow \{A_k^c \text{ does not occur at most finitely many } k\}$ has probability 1.

$\Rightarrow$ With probability 1, $\lim_{k \rightarrow \infty} \frac{N(0, k^2)}{k^2} = \lambda$.

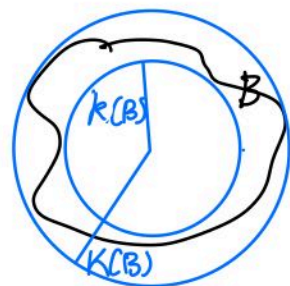
taking $k = \lfloor t^{1/2} \rfloor$ and use sandwich thm gives that $\lim_{t \rightarrow \infty} \frac{N(0, t)}{t} = \lambda$. □

This proof can be extended to general Poisson process, like $S = \mathbb{R}^d$ case,

for A_k : sphere of radius k , $\Rightarrow |A_k| = |A_1| \cdot k^d$

$\Rightarrow$ By Borel-Cantelli & Above proof, $\lim_{k \rightarrow \infty} \frac{N(A_k)}{|A_k|} = \lambda$

$\Rightarrow$ let $k(B) := \sup_k \{A_k \subset B\}$ $K(B) = \inf_k \{B \subset A_k\}$



$$\Rightarrow N(A_{k(B)}) \leq N(B) \leq N(A_{K(B)}) \quad \& \quad |A_{k(B)}| \leq |B| \leq |A_{K(B)}|$$

$$\Rightarrow \lim_{|B|} \frac{N(B)}{|B|} = \lambda \text{ a.s. as long as } k(B) \rightarrow \infty, K(B)/k(B) \rightarrow 1.$$

③ Non-homogeneous process.

Now for a non-homogeneous Poisson process π on $\mathbb{R}$ with mem measure μ .

$$\text{Let } M(t) = \begin{cases} \mu(0, t] & t \geq 0 : \text{continuous and increasing} \\ \mu(-t, 0] & t < 0 \end{cases}$$

By Mapping thm, $\pi' := M(\pi)$ is a Poisson process

$$\text{and for } x = M(t), \mu^*(0, x] = \mu^*(0, M(t)] = \mu(0, t] = M(t) = x.$$

$\Rightarrow \pi'$ has a rate 1.

Only possible ambiguity: M has a flat section $\leftarrow$ only at cble pts.

But Unfortunately, The intervals of π are not independent.

$$\begin{aligned} \text{Compute } P(Y_2 - Y_1 \leq t | Y_1 = y_1) &= P(N(y_1, y_1 + t) \geq 1 | Y_1 = y_1) \\ &= 1 - e^{-\int_{y_1}^{y_1+t} \lambda(s) ds} : \text{fn of } y_1 \end{aligned}$$

$\Rightarrow Y_2 - Y_1$ and Y_1 are not independent,

But as we proved in 4.2, the Law of large number holds,

Example. Π_2 : Poisson distribution on plane with const. rate λ .

let $\pi: \{ \|X\| \mid X \in \pi \} : \text{forms a Poisson process on } (0, \infty)$

with rate $\lambda(x) = 2\pi\lambda x$.

$\Rightarrow$ with $M(x) = \pi\lambda x^2$, $\{M(X) \mid X \in \pi\}$: Poisson process of rate 1.

let if $\pi_M = \{Y_1, Y_2, Y_3, \dots\} \Rightarrow$ By the Strong law of the large numbers,

$$\lim_{n \rightarrow \infty} \frac{Y_n}{n} = 1 \text{ a.s.}$$

Then for $N_\pi[0, t]$: # of points of π in $(0, t]$

is equal to $N_{\pi_M}[0, M(t)]$: # of points of Y_n in $(0, M(t)]$

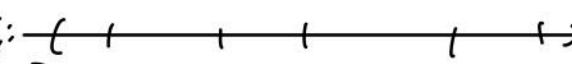
$$\Rightarrow \lim_{t \rightarrow \infty} \frac{N_\pi[0, t]}{M(t)} = 1 \text{ a.s.}$$

Final example: If the rate fn $\lambda(x)$ satisfies $\int_0^\infty \lambda(x) dx = \infty$, $\int_t^\infty \lambda(x) dx < \infty \forall t > 0$

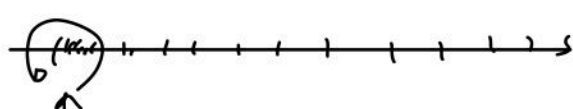
In this case, let $L(t) = \mu(t, \infty) = \int_t^\infty \lambda(x) dx$.

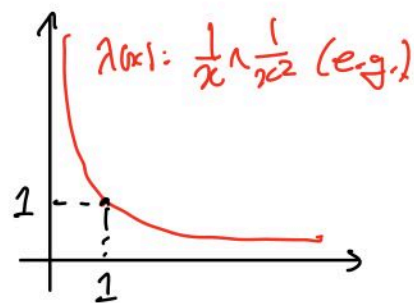
$\Rightarrow L$: conti. decreasing, with $L(0) = \infty$, $L(\infty) = 0$

By Mapping then,

$L(\pi)$: Poisson process of rate 1, π : 

By Inverting the mapping, $N_\pi(0, t] = N_{\pi_M}(0, L(t)) \sim P(L(t))$

$\Rightarrow \pi$: 
↑ accumulated at origin.



From the Law of the lar num, we have $\lim_{t \rightarrow \infty} \frac{N(t, \infty)}{L(t)} = 1 \text{ a.s.}$

③ Queues.

Now coming back to the const. rate Poisson process, Consider the system.

① The customers arrive according to a Π_α Poisson process with rate α .

③ Each customer needs a service that take time,

the service times are independent among the customers,

and follows the $\text{Exp}(\beta)$

$\Rightarrow$ By Interval theorem, we may regard the point that each customer finishes their task and leaves, is a point of Π_β .

(Note that the converse - At Every point of Π_β , customer leaves - is not true)

This is a Queueing system $M/M/1$. Each M indicates that

The arrival time / The leaving time is Markovian Process, and 1 indicates

That there are only one server.

How can we analyze this system?

let $P_n(t) := \mathbb{P}(\text{There are } n \text{ customers in queue at time } t)$

We will make the ODE that $P_n(t)$ follows,

for $h \ll 1$, on time interval $(t, t+h)$, there are $\begin{cases} \text{arrival with prob. } \alpha h + o(h) \\ \text{departure " " } \beta h + o(h) \text{ (if } n \geq 1) \end{cases}$

$$(\because \mathbb{P}(N(t, t+h] \geq 1) = 1 - e^{-\lambda h} = \lambda h + o(h))$$

$$P_n(t+h) = P_n(t)(1-\alpha h - \beta h + o(h)) + P_{n-1}(t)(\alpha h + o(h)) + P_{n+1}(t)(\beta h + o(h)) \quad (n \geq 1)$$

$$\Rightarrow \begin{cases} \frac{d}{dt} P_n(t) = -(\alpha + \beta) P_n(t) + \alpha P_{n-1}(t) + \beta P_{n+1}(t) & (n > 0) \\ \frac{d}{dt} P_0(t) = -\alpha P_0(t) + \beta P_1(t) & (n = 0) \end{cases}$$

Assume Queue is empty when $t=0$. $\sim P_n(0) = \delta_{n0} = \begin{cases} 1 & n=0 \\ 0 & n \neq 0 \end{cases}$

IDEA $\rightarrow$ Laplace transform. $\mathcal{L}\{P_n(t)\}(\lambda) = \int_0^\infty P_n(t) e^{-\lambda t} dt$

let $r_n = \mathcal{L}\{P_n(t)\}$

Then r_n satisfies

$$\begin{aligned} -(\alpha + \beta) r_n(\lambda) + \alpha r_{n-1}(\lambda) + \beta r_{n+1}(\lambda) &= \int_0^\infty \frac{d}{dt} P_n(t) e^{-\lambda t} dt \quad (\text{Integration By parts}) \\ &= \int_0^\infty \lambda P_n(t) e^{-\lambda t} dt = \lambda r_n(\lambda) \end{aligned}$$

$$\Rightarrow \beta r_{n+1}^2 - (\alpha + \beta + \lambda) r_n + \alpha r_{n-1} = 0$$

let ξ, η : sol. of $\beta x^2 - (\alpha + \beta + \lambda)x + \alpha = 0$.

$$\Rightarrow r_n = C_1 \cdot \xi^n + C_2 \cdot \eta^n$$

Note: $\sum r_n(\lambda) = \int_0^\infty \sum P_n(t) e^{-\lambda t} dt = \int_0^\infty e^{-\lambda t} dt = \frac{1}{\lambda} < \infty \dots (*)$

$\Rightarrow r_n$ are bounded, Hence exactly one of ξ & η lies $|x| < 1$. let it ξ .

$$\Rightarrow r_n = C_1 \cdot \xi^n. \text{ by } (*), C_1 \cdot \frac{1}{1-\xi} = \frac{1}{\lambda}. \Rightarrow C_1 = \frac{(1-\xi)}{\lambda}, r_n(\lambda) = \frac{1}{\lambda} (1-\xi) \xi^n$$

where $\xi = \frac{1}{2\beta} \left\{ \alpha + \beta + \lambda - \sqrt{(\alpha + \beta + \lambda)^2 - 4\alpha\beta} \right\}$

Similar form with the $\mathcal{L}\{J_n\}$ $\leftarrow$ Not that interested.

In particular, $r_0(\lambda) = \frac{1-\xi}{\lambda} = \frac{-\alpha+\beta-\lambda + \sqrt{(\alpha+\beta+\lambda)^2 - 4\alpha\beta}}{2\beta\lambda}$

$P_0(t) = \sum_{0 \leq r \leq s} \frac{s-r+1}{s+1} \cdot \pi_r(\alpha t) \cdot \pi_s(\beta t)$, where $\pi_r(\alpha t) = \frac{(\alpha t)^r e^{-\alpha t}}{r!}$
 $= P(N_{\pi_\alpha}(0, t] = r)$

pf 1) Claim $\mathcal{L}\{\text{RHS}\} = \frac{1-\xi}{\lambda} = r_0(\lambda) \Rightarrow P_0(t) = \text{RHS}$.

$$\mathcal{L}\{\text{RHS}\}(\lambda) = \int_0^\infty \sum_{0 \leq r \leq s} \frac{s-r+1}{s+1} \cdot \frac{(\alpha t)^r e^{-\alpha t}}{r!} \cdot \frac{(\beta t)^s e^{-\beta t}}{s!} e^{-\lambda t} dt$$

$$= \int_0^\infty \sum_{0 \leq r \leq s} \frac{(s-r+1) \alpha^r \beta^s}{r! (s+1)!} \cdot \underbrace{t^{r+s}}_{\mathcal{L}\{t^{r+s}\}} \cdot e^{-(\alpha+\beta+\lambda)t} dt$$

$$= \sum_{0 \leq r \leq s} \frac{(s-r+1) \alpha^r \beta^s}{r! (s+1)!} \frac{\lambda}{(\alpha+\beta+\lambda)^{r+s+1}}$$

($\because \mathcal{L}\{t^n\}(\lambda) = \frac{n!}{\lambda^{n+1}}$)

let $u = \frac{\alpha}{\alpha+\beta+\lambda}$, $v = \frac{\beta}{\alpha+\beta+\lambda}$

$$\Rightarrow \lambda \cdot \mathcal{L}\{\text{RHS}\}(\lambda) = \sum_{\substack{0 \leq r \leq s \\ a_{r,s}}} \frac{(s-r+1)}{r! (s+1)!} u^r v^s (1-u-v)$$

Coefficient of $u^r v^s = 0$ unless $r=s=0$ or $s=r-1$

① $r=2 \geq s \rightarrow 0$

② $s=r-1$: $u^r v^{r-1}$ only appears at $a_{r,r-1} = \frac{1 \cdot (2r-2)!}{(r-1)! r!} \neq 0$

③ $s=r$ & $r \neq 0$: $u^r v^s$ appears $a_{r,r} - a_{r-1,r} = \frac{2r!}{r! (r+1)!} - \frac{2r(2r-1)!}{(r-1)! (r+1)!} = 0$

④ $s=r$ & $r=0$: $u^r v^s$: $a_{0,0} = \frac{1 \cdot 0!}{0! 1!} = 1$

submitted
in
seminar

$$\begin{aligned}
 \textcircled{5} \quad s \geq r+1 \text{ \& } r \neq 0 : u^r v^s : \text{ appears } a_{r,s} - a_{r-1,s} - a_{r,s-1} \\
 &= \frac{(s-r+1)(r+s)!}{r!(s+1)!} - \frac{(s-r+2)(r+s-1)!}{(r-1)!(s+1)!} - \frac{(s-r)(r+s-1)!}{r!s!} \\
 &= \frac{(r+s-1)!}{(r-1)!s!} \left(\frac{(s-r+1)(r+s)}{s(s+1)} - \frac{s-r+2}{s+1} - \frac{s-r}{r} \right) \\
 &= \frac{r+s}{r} - \frac{r+s}{s+1} - \frac{s-r+2}{s+1} - \frac{s-r}{r} = 2 - 2 = 0
 \end{aligned}$$

$$\textcircled{6} \quad s \geq r+1 \text{ \& } r=0 : u^r v^s : a_{0,s} - a_{0,s-1} = \frac{(s+1)s!}{(s+1)!} - \frac{s \cdot (s-1)!}{s!} = 0.$$

$$\begin{aligned}
 \Rightarrow \lambda \cdot \mathbb{L}\{\text{RHS}\}(\lambda) &= 1 - \sum_{r=1}^{\infty} \frac{(2r-2)!}{(r-1)!r!} u^r v^{r-1} \\
 &= 1 - \frac{1}{2v} (1 - \sqrt{1-4uv}) \\
 &= 1 - \frac{\alpha+\beta+\lambda}{2\beta} \cdot \left(\frac{\alpha+\beta+\lambda - \sqrt{(\alpha+\beta+\lambda)^2 - 4\alpha\beta}}{\alpha+\beta+\lambda} \right) \\
 &= 1 - \varphi \quad \therefore \mathbb{L}\{\text{RHS}\} = r_0, \quad p_0(t) = \text{RHS}. \quad \square
 \end{aligned}$$

pf2) W.T.S: $p_0(t) = \sum_{0 \leq r \leq s} \frac{s-r+1}{s+1} \underbrace{\pi_r(\alpha t)}_{\substack{\uparrow \\ \# \text{ arrivals} \\ \text{in } (0,t]}} \underbrace{\pi_s(\beta t)}_{\substack{\uparrow \\ \# \text{ possible departure} \\ \text{in } (0,t]}}$

$$p_0(t) = \sum_{(r,s)} b_{r,s} \cdot \pi_r(\alpha t) \pi_s(\beta t) \quad (\text{Law of total prob.})$$

$\rightarrow b_{r,s}$ = Probability that queue is empty at the t

where r arrivals and s (possible) departure was made.

Claim $b_{r,s} := \begin{cases} \frac{s-r+1}{s+1} & 0 \leq r \leq s \\ 0 & \text{o/w} \end{cases}$

pt) $b_{rs} = 0$ if $r > s$ is easy.

for $r \leq s$ case, calculating b_{rs} is equivalent to the following problem

: Counting the # of minimal lattice path of $(0,0)$ to (s,r)

which does not touches the line $y = x - (s-r)$

Why? correspond $\rightarrow$ to departure, $\uparrow$ to arrival.

for every point, (# of departure - # arrival) should be less than $s-r$.

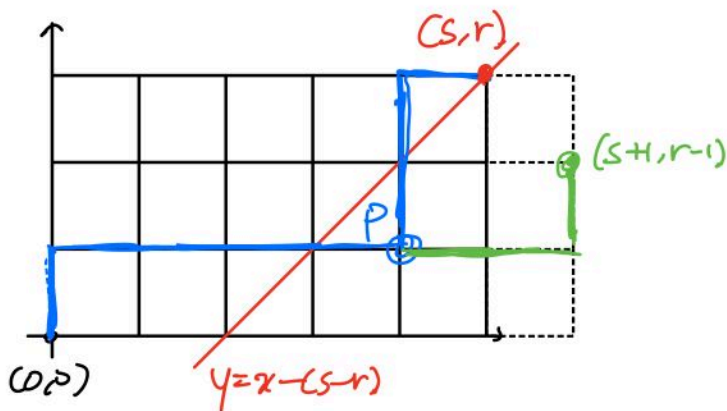
If not, the # of the (possible) departure that the actual departure not happened, is greater than $s-r$.

Now, If a path across the line $y = x - s + r$,

let P : first point that touches $y = x - (s-r) - 1$

$\Rightarrow$ reflect the path $P \rightarrow (s,r)$ w.r.t $y = x - (s-r) - 1$

$\Rightarrow$ we obtain the path from $(0,0)$ to $(s+1, r-1)$



$\therefore$ # of minimal lattice path of $(0,0)$ to (s,r)

which does not touches the line $y = x - (s-r)$

$$= \binom{s+r}{s} - \binom{s+r}{s+1}$$

$$\text{and } b_{rs} = \left(\binom{s+r}{s} - \binom{s+r}{s+1} \right) / \binom{s+r}{s} = 1 - \frac{r}{s+1} = \frac{s-r+1}{s+1}$$

$$\text{Therefore, } p_0(t) = \sum_{0 \leq r \leq s} \frac{s-r+1}{s+1} \pi_r(\alpha t) \pi_s(\beta t)$$



4) Bartlett's theorem.

Now there are k -servers,

Then the equations for $P_n(t)$ are easily changed as

$$\begin{cases} P_n'(t) = P_n(t)(1 - (\alpha + k\beta)h + o(h)) + P_{n-1}(t)(\alpha h + o(h)) + P_{n+1}(t)(\beta k h + o(h)) & (n \geq k) \\ P_n'(t) = P_n(t)(1 - \alpha h - \beta n h + o(h)) + P_{n-1}(t)(\alpha h + o(h)) + P_{n+1}(t)(\beta(n+1)h + o(h)) & (n < k) \end{cases}$$

Now we will analyze the equilibrium of this system, so let $P_n(t) \equiv P_n : \text{const.}$

$$\Rightarrow (\alpha + \beta(k \wedge n)) P_n = \alpha P_{n-1} + \beta(k \wedge (n+1)) P_{n+1}$$

$$\Rightarrow \alpha P_n - \beta(k \wedge (n+1)) P_{n+1} = \alpha P_{n-1} - \beta(k \wedge n) P_n$$

$$\Rightarrow \alpha P_n - \beta(k \wedge n) P_{n+1} = \text{const} = 0 \quad (\because P_n \rightarrow 0 \text{ as } n \rightarrow \infty)$$

$$\Rightarrow P_n / P_{n+1} = \alpha / \beta \cdot \frac{1}{(n \wedge k)} \Rightarrow P_n = \begin{cases} P_0 \cdot \frac{(\frac{\alpha}{\beta})^n}{n!} & (n \leq k) \\ P_0 \cdot \frac{(\frac{\alpha}{\beta})^n}{k^{n-k} \cdot k!} & (n > k) \end{cases}$$

and from $\sum P_n = 1$, $1 = P_0 \cdot \left(\sum_{n=0}^k \left(\frac{\alpha}{\beta}\right)^n \frac{1}{n!} + \sum_{n=k+1}^{\infty} \frac{(\frac{\alpha}{\beta})^n}{k! k^{n-k}} \right)$

Note that the second term converges iff $|\frac{\alpha}{\beta k}| < 1$.

otherwise, There are no equilibrium (or stationary solution)

↳ in this case, one can show that, for each $n \in \mathbb{N}$,

$P_n(t) \rightarrow 0$ as $t \rightarrow \infty$ for all initial distribution of $\{P_n(0)\}_{n \in \mathbb{N}}$.

Now let $k \rightarrow \infty$ infinite number of servers,

then the equilibrium $P_0 = e^{-\frac{\alpha}{\beta}}$, $P_n = \frac{(\frac{\alpha}{\beta})^n e^{-\frac{\alpha}{\beta}}}{n!} = \pi_{\alpha/\beta}(n)$

Moreover, the differential equation takes the form

$$\frac{dp_n}{dt} = -(\alpha + \beta n) p_n + \alpha p_{n-1} + \beta (n+1) p_{n+1}$$

We put the ansatz $p_n(t) = \pi_n(\mu(t))$ for appropriate fn μ .

$$\text{then } \frac{d\mu}{dt} (\pi_{n-1} - \pi_n) = -\alpha \pi_n - \beta \mu \pi_{n-1} + \alpha \pi_{n-1} + \beta \mu \pi_n \quad (\because n \pi_n = \mu \pi_{n-1})$$

$$\Rightarrow \frac{d\mu}{dt} = \alpha - \beta \mu. \quad \text{then } \mu(t) = \frac{\alpha}{\beta} + \left(\mu(0) - \frac{\alpha}{\beta}\right) e^{-\beta t} \quad \dots (**)$$

Hence, if (# of cust.) follows Poisson distribution with mean $\mu(0)$ at time $t=0$,

then it follows Poisson distribution with mean $\mu(t) \quad \forall t > 0$.

Intuitive Explanation: If $K=\infty \rightarrow$ The customers does not interfere each other.

Bartlett's theorem generalizes this result into $\pi_{\alpha(t)}$, not constant rate α .

Thm (Bartlett)

Customers arrives according to Poisson process of rate $\alpha(t)$.

They moves at random around the system in such a way their trajectories are independent.

Let E : subset of a system,

$p(s,t)$: probability that the customer who arrives at time s being in E

at the subsequent time t .

Then, the # of customers in E at time t has a Poisson distribution, with mean $\mu(t) = \int_{-\infty}^t \alpha(s) p(s, t) ds$

The result is quite intuitive, the proof is passed to chapter 5.

Now let's apply this then into $M/M/\infty$ case.

Suppose $\alpha(t) = \begin{cases} 0 & t < 0 \\ \alpha & t \geq 0 \end{cases}$, and E : queue.

$\Rightarrow p(s, t) = q(t-s)$ where $q(k) :=$ probability that the customer spends more than time k in the queue.

Since $q(k) = P(\text{service time} \geq k) = \int_k^{\infty} \beta e^{-\beta t} dt = e^{-\beta k}$.

By Bartlett, (# of customers in queue at time t) $\sim P(\mu(t))$

where $\mu(t) = \int_{-\infty}^t \alpha(s) p(s, t) ds = \int_0^t \alpha e^{-\beta(t-s)} ds = \frac{\alpha}{\beta} (1 - e^{-\beta t})$

this coincides with (**) with $\mu(0) = 0$.

More generally, The queue $M/G/\infty$ (service time has general distribution)

By Bartlett, (# of customers in queue) $\sim P(\mu(t))$

where $\mu(t) = \alpha \cdot \int_0^t (1 - B(u)) du$

($B(u)$: distribution cdf)

End.